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Cairo University, Faculty of Engineering Department
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ELC 4028- Artificial Neural Networks and its Applications

Assignment 1

Classical Machine Learning Methods

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Part 1: Classical ML modeling

This part presents the implementation and evaluation of classical machine learning methods for handwritten digit classification using the **Reduced MNIST** dataset — a balanced subset of the standard MNIST benchmark. The study investigates three distinct feature extraction strategies and two classifier families, comparing their performance systematically across all combinations.

Dataset — Reduced MNIST

The Reduced MNIST dataset is a balanced subset of the MNIST handwritten digit database.
Each of the 10 digits classes (0–9) is represented by an equal number of samples:

- Training set: 1,000 images per class → 10,000 total training images.
- Test set: 200 images per class → 2,000 total test images.
- Each image: 28 × 28 grayscale pixels (784 raw features).

Images were drawn uniformly at random (without replacement) from the full MNIST splits, then globally shuffled to ensure class independence. The figure below shows one sample from each class.

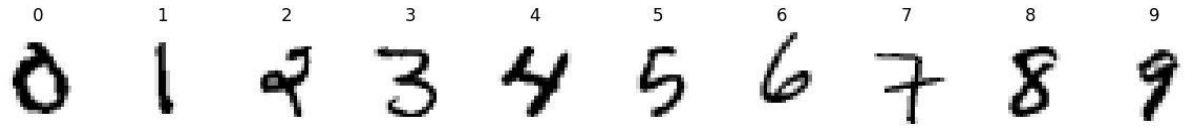


Figure 1: One sample from each digit class (0–9) in the Reduced MNIST training set.

Pipeline 2: Manual Seed Labelling with Augmentation and Self-Training Refinement

1. Overview

The core idea in this pipeline is to start with a small, human-labelled trusted set, expand it through data augmentation, and iteratively grow the labelled pool by combining active learning on uncertain boundary examples with self-training on high-confidence predictions.

The baseline reference is full manual labelling: 10,000 images × 10 seconds = 100,000 seconds ≈ 27.8 hours. Pipeline 2 achieves a labelling accuracy of 97.43% using only 740 manually labelled images (300 seed + 440 boundary), corresponding to 7,400 seconds ≈ 2.06 hours of human effort which is a reduction of approximately 93% in manual time.

2. Pipeline Block Diagram

Figure 1 shows the full pipeline steps as a block diagram, including the exact parameter values used in the implementation.

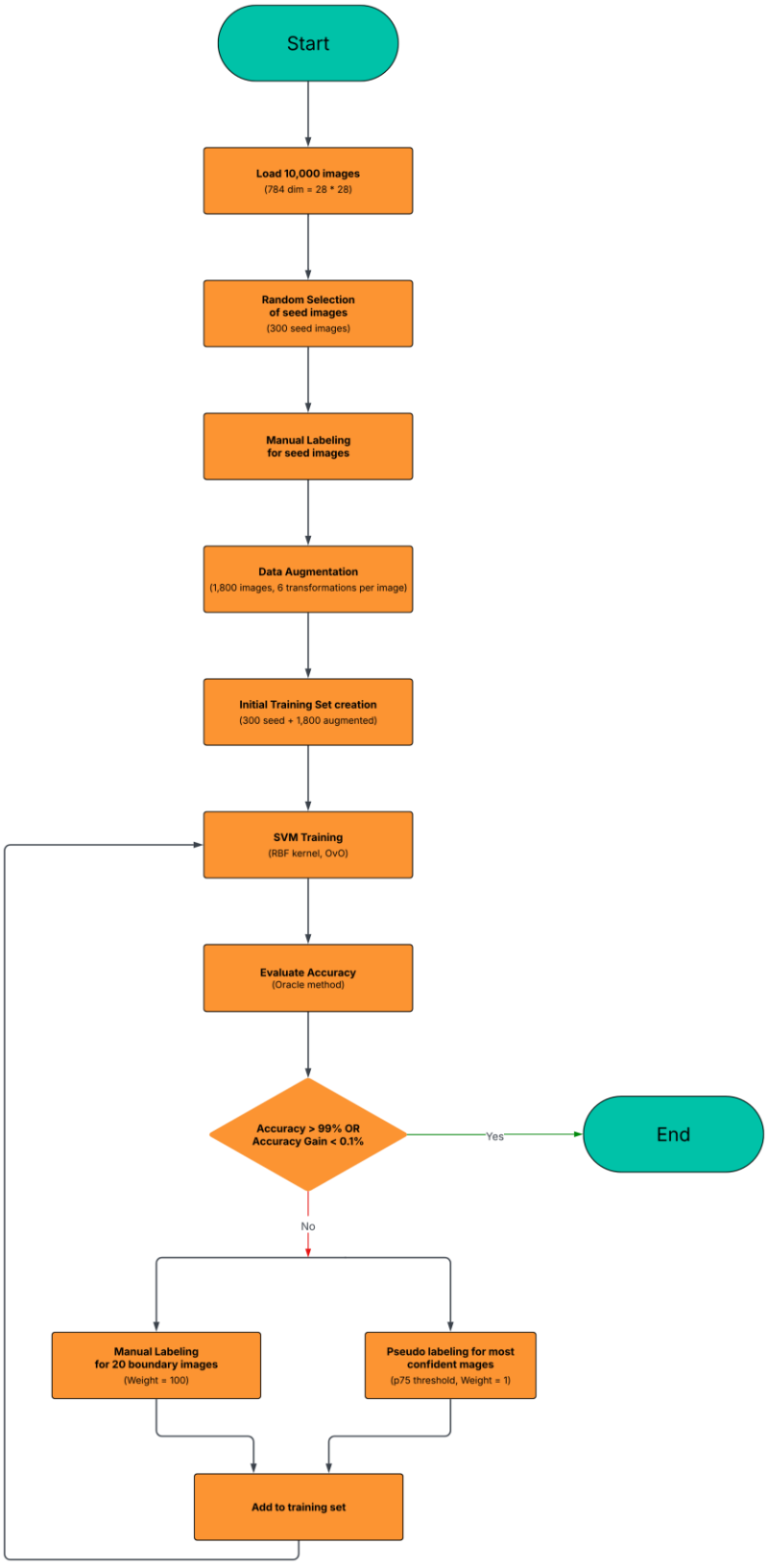


Figure 2: Pipeline 2 block diagram with parameter values.

2. Summary Table

Metric	Value
Final Labeling Accuracy (Oracle)	97.43%
Number of iterations	22
Total number of images manually labelled	740 (300 seed + 20 *22 boundary)
Total number of pseudo-labelled images incorporated	3528
Total estimated manual time	123.3 min

Table 1: Summary for required parameters.

Manual Labelling Time

A central metric for comparing labelling pipelines is the total manual effort required. Table 2 summarizes the manual time breakdown for Pipeline 2, alongside the full-manual baseline.

Activity	Images	Time (s)	Time (min)
Seed labelling (Step 1)	300	3,000	50.0
Boundary labelling (22 iter × 20)	440	4,400	73.3
Total — Pipeline 2	740	7,400	123.3
Full manual baseline	10,000	100,000	1,667 (27.8 h)

Table 2: Manual labelling time comparison.

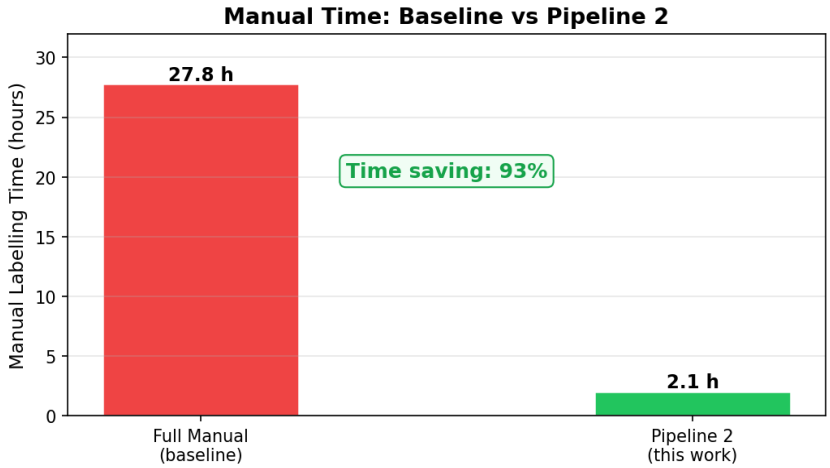


Figure 3: Manual labelling time: full manual baseline vs Pipeline 2.

Pipeline 2 required 740 manually labelled images (7,400 seconds ≈ 2.06 hours), compared to 100,000 seconds ≈ 27.8 hours for the full manual baseline. This represents a 92.6% reduction in human effort.

Iteration Log

Table 3 presents the complete per-iteration log of accuracy and corresponding train set size.

Iter	Accuracy (%)	Pseudo Added	Pseudo Rejected	Train Set Size
1	89.22	458	7395	2578
2	90.20	403	7376	3001
3	91.51	404	7343	3425
4	92.54	399	7291	3844
5	93.19	376	7206	4240
6	93.42	317	7064	4577
7	93.78	227	6896	4824
8	94.29	163	6760	5007
9	94.62	121	6643	5148
10	95.03	99	6533	5267
11	95.43	74	6439	5361
12	95.79	49	6370	5430
13	95.98	72	6278	5522
14	96.18	64	6194	5606
15	96.52	26	6148	5652
16	96.71	43	6085	5715
17	96.86	51	6014	5786
18	97.00	47	5947	5853
19	97.12	37	5890	5910
20	97.28	37	5833	5967
21	97.36	36	5777	6023
22	97.43	25	5732	6068

Table 3: Iteration-by-iteration results log.

3. Implementation Details

Step 1 — Load Full Dataset

All 10,000 images from Indian_Digits_Train were loaded from disk. Each 28×28 grayscale image was flattened to a 784-dimensional pixel vector and normalized to the range [0, 1]. Files were sorted numerically to ensure consistent row-to-filename mapping throughout the pipeline.

Step 2a — Random Seed Sampling and Manual Labelling

A random subset of 300 images was selected from the unlabelled dataset , ensuring no duplicates. Since the dataset order is unknown and not sequential by class, the sample was drawn uniformly at random from all 10,000 images, and manually assigned the correct digit label (0–9) to each of the 300 images.

The resulting seed label distribution was:

Class 0	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9
35	34	32	27	27	22	33	25	33	32

Table 4: Seed label distribution across 10 classes.

Estimated manual time for this step: **300 images × 10 s = 3,000 s ≈ 50 minutes.**

Step 2b — Data Augmentation

To compensate for the small seed size and improve classifier robustness, each of the 300 seed images was augmented by applying 6 distinct transformations:

- Rotation +5°
- Rotation −5°
- Additive Gaussian noise (mean = 0, variance = 0.002)
- Horizontal shift by +2 pixels
- Vertical shift by +2 pixels
- Horizontal shift by −2 pixels

Each transformation produces one additional image, yielding 300 × 6 = 1,800 augmented images. Augmented images inherit the label of their source image and are assigned a sample weight of 1, reflecting their synthetic origin.

Step 2c — Initial Training Set Assembly

The initial training set combines seeds and augmented images as follows:

Source	Count	Weight	Rationale
Seed images (human-labelled)	300	100	Trusted ground truth
Augmented copies	1,800	1	Synthetic, lower trust
Total	2,100	—	

Table 5: Initial training set composition.

The weight ratio of 100:1 between seed and augmented images ensures that the SVM strongly respects human-verified labels during training. This is equivalent to each seed image having 100 times the influence on the decision boundary compared to a single augmented copy.

Step 3 — Initial SVM Training (SVM-1)

A multi-class Support Vector Machine (SVM) was trained on the initial training set using a one-vs-one (OvO) scheme. The following configuration was used:

- Kernel function: Radial Basis Function (RBF)
- Kernel scale: estimated automatically by MATLAB
- Box constraint (C): 10
- Feature standardization: enabled (Standardize = true)
- Multi-class scheme: one-vs-one (OvO), 45 binary classifiers for 10 classes

After training, predictions and decision function scores were computed for all 10,000 images. The confidence margin for each image was defined as the difference between the highest and second-highest decision function score across all binary classifiers. SVM-1 achieved an initial labelling accuracy of **89.2%** on the full dataset, with a roughly balanced prediction distribution across all 10 classes.

Step 4 — Active Labelling of Boundary Images

The pipeline then enters an iterative loop that alternates between human labelling of uncertain examples (active learning) and automatic labelling of confident predictions (self-training). The loop runs until either the target accuracy of 99% is reached, or the improvement between consecutive iterations falls below the convergence threshold of 0.01% for two consecutive iterations.

In each iteration, the 20 unlabeled images with the smallest confidence margin are identified. These images sit closest to the SVM's decision boundaries and are most likely to be mislabeled by the current model. They were manually labeled at 10 seconds per image, and added to the training set with weight = 100.

Manual time per iteration: **20 images × 10 s = 200 seconds.**

Step 5 — Self-Training

In addition to the human-labelled boundary images, the training set is expanded by selecting high-confidence pseudo-labelled images. Following the assignment specification, the selection criterion is as follows:

1. Compute a global threshold: the 75th percentile of all margin scores across the current iteration.
2. For each class (0–9), identify unlabeled images that are: (a) predicted as that class, (b) not already in the training set, and (c) have a margin above the global threshold.
3. From the qualifying candidates, select the top-50 images by margin score given that their margin is above the threshold.
4. Assign these images pseudo-labels equal to the current SVM prediction, with weight = 1.

This adds up to 500 pseudo-labelled images per iteration (50 per class × 10 classes), subject to pool availability and the threshold filter.

Step 6 — Retrain and Evaluate

The SVM is retrained on the updated training set (seeds + augmented + boundary + pseudo-labelled). Predictions and margins are recomputed for all 10,000 images, and the oracle function check_accuracy is called to obtain the exact labelling accuracy.

Step 7 — Stopping Conditions

The loop terminates when any of the following conditions are met:

- Labelling accuracy reaches or exceeds 99% (target threshold).
- The gain in accuracy between consecutive iterations is below 0.1% (conv_delta = 0.001) for two consecutive iterations, indicating convergence.

4. Results

4.1 Accuracy Progression

The pipeline ran for 22 iterations before converging. Figure 3 shows the labelling accuracy at each iteration, starting from 89.22% after the first refinement iteration (SVM-1 baseline: 89.2%) and reaching 97.43% at convergence. The 99% target was not reached within 22 iterations; convergence was triggered because the accuracy gain fell below 0.01% for two consecutive iterations (iterations 21 and 22).

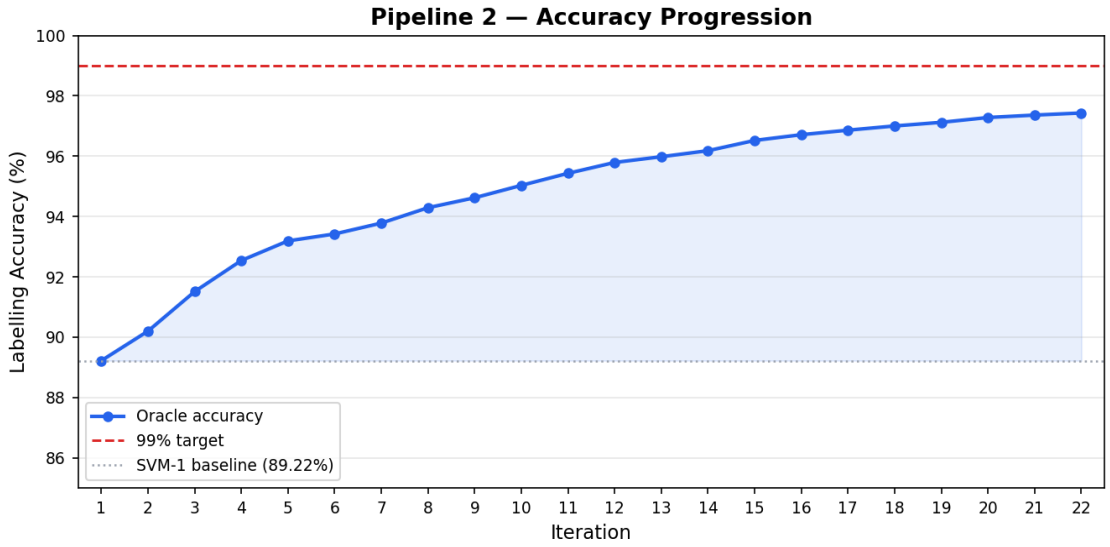


Figure 4: Labelling accuracy per iteration.

The accuracy improvement follows a characteristic self-training curve: relatively fast gains in early iterations (approximately 1–2% per iteration) as the most confidently predicted images are incorporated, followed by diminishing returns as the remaining unlabeled images become increasingly ambiguous. The average gain per iteration across all 22 iterations was 0.37%.

4.2 Self-Training Activity

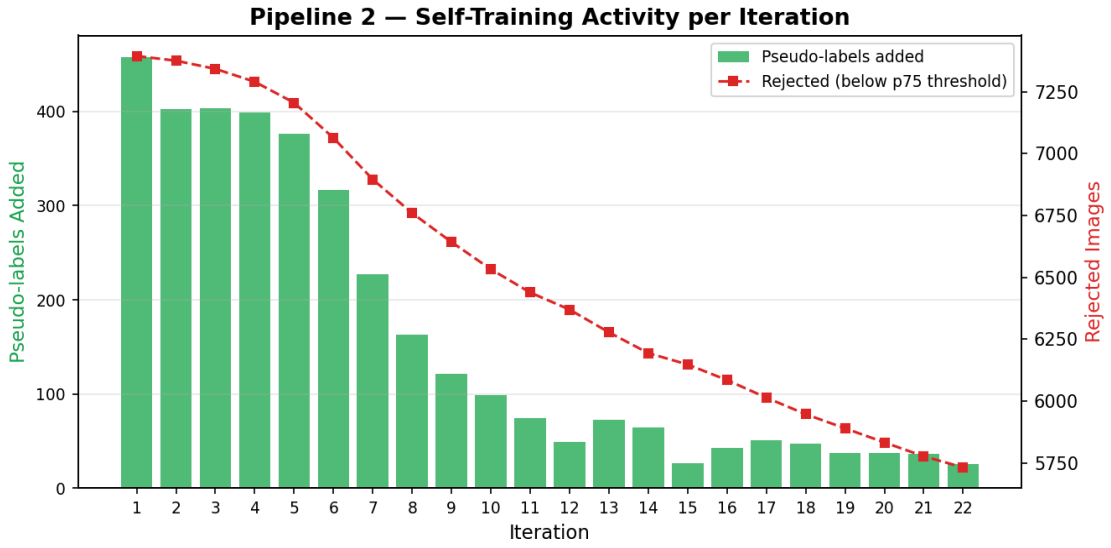


Figure 5: Pseudo-labels added (green bars) and rejected by the global p75 threshold (red line) per iteration.

Figure 4 shows a clear trend: the number of pseudo-labels added per iteration decreases sharply after iteration 1 (from 458 down to 25 by iteration 22). This reflects the tightening of the global 75th percentile threshold as the SVM becomes more confident overall — the threshold rises as more images accumulate high margins, leaving fewer candidates above it. Simultaneously, the number of rejected images decreases gradually as the unlabeled pool shrinks.

4.2 Training Set Growth

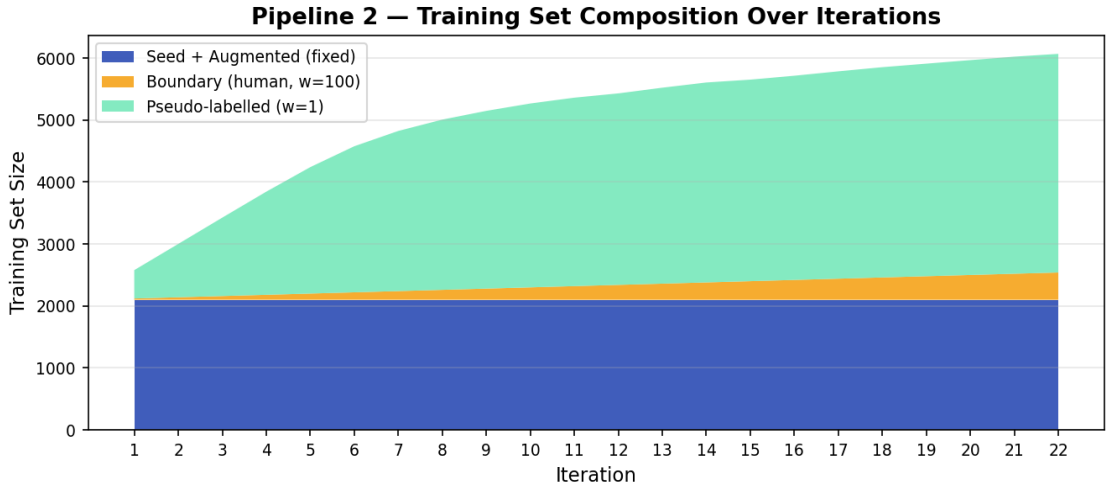


Figure 6: Training set composition over 22 iterations

The training set grew from 2,100 samples at initialization to approximately 6068 samples by iteration 22. The fixed layer (seed + augmented) always constitutes the largest portion, providing a stable trusted foundation. Boundary images accumulated steadily at 20 per iteration. Pseudo-labelled images grew rapidly early on and then stablize as fewer confident unlabeled images remained.

7. Code Parameters Summary

Table 6 provides a summary of all parameters used in the code for this pipeline.

Parameter	Value
Feature representation	Raw pixels (784-dim)
Seed set size	300 images
Augmentations per image	6
Seed weight	100
Augmented weight	1
SVM kernel	RBF
KernelScale	auto
BoxConstraint (C)	10
N_uncertain (boundary)	20
N_per_class (pseudo)	50
Pseudo threshold	Global 75th pct
Convergence delta	0.001 (0.1%)
Max iterations	50

Table 6 — Pipeline 2 code parameters.